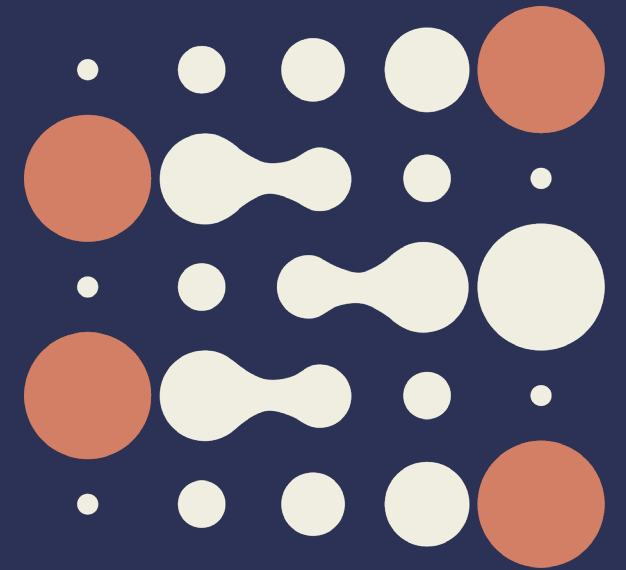


Evidence synthesis and integration for cancer hazard identification in *IARC Monographs* programme: role of genomic epidemiology

Cancer Genomic Epidemiology course
Wellcome Connecting Science
18 February 2026

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International Agency for Research on Cancer
Evidence Synthesis and Classification Branch
Lyon, France

International Agency
for Research on Cancer



Presentation overview

- IARC and the *Monographs* – background
- The *Monographs* Preamble – modernized evidence synthesis and integration across evidence streams
- Priorities for *IARC Monographs* evaluation
- Genomic epidemiology in the *IARC Monographs*



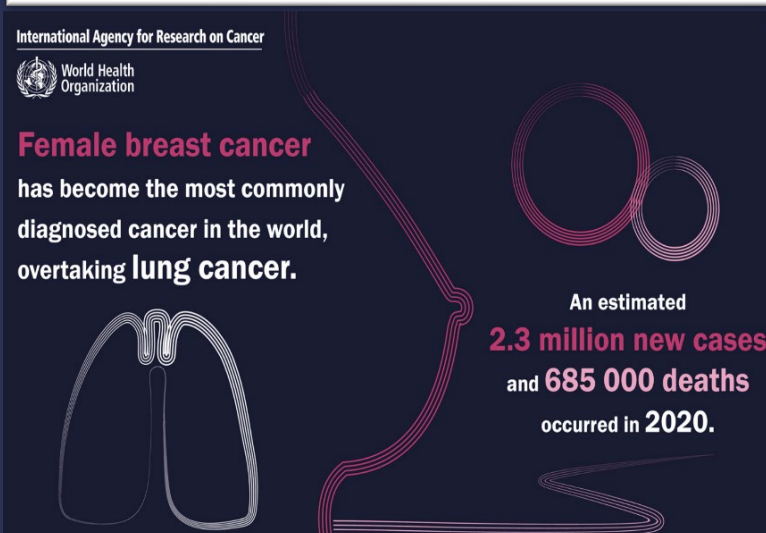
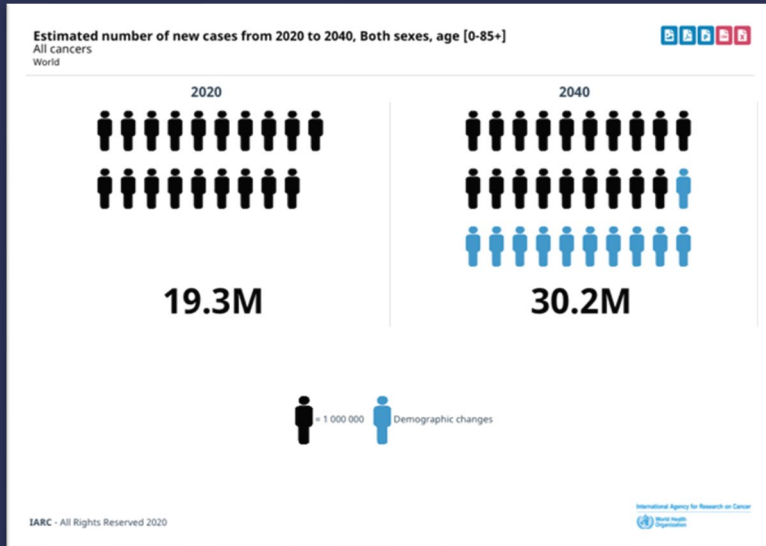
Presentation overview

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Cancer: the global burden

- Epidemiologic transition → rising global burden
 - Changing demographics
 - Shift in carcinogenic exposures to LMICs
- Prevention is the most effective response
- The first step in cancer prevention is to identify
 - **causes** of human cancer (*IARC Monographs*) and
 - **what prevents cancer** (*IARC Handbooks*)



<https://monographs.iarc.who.int>

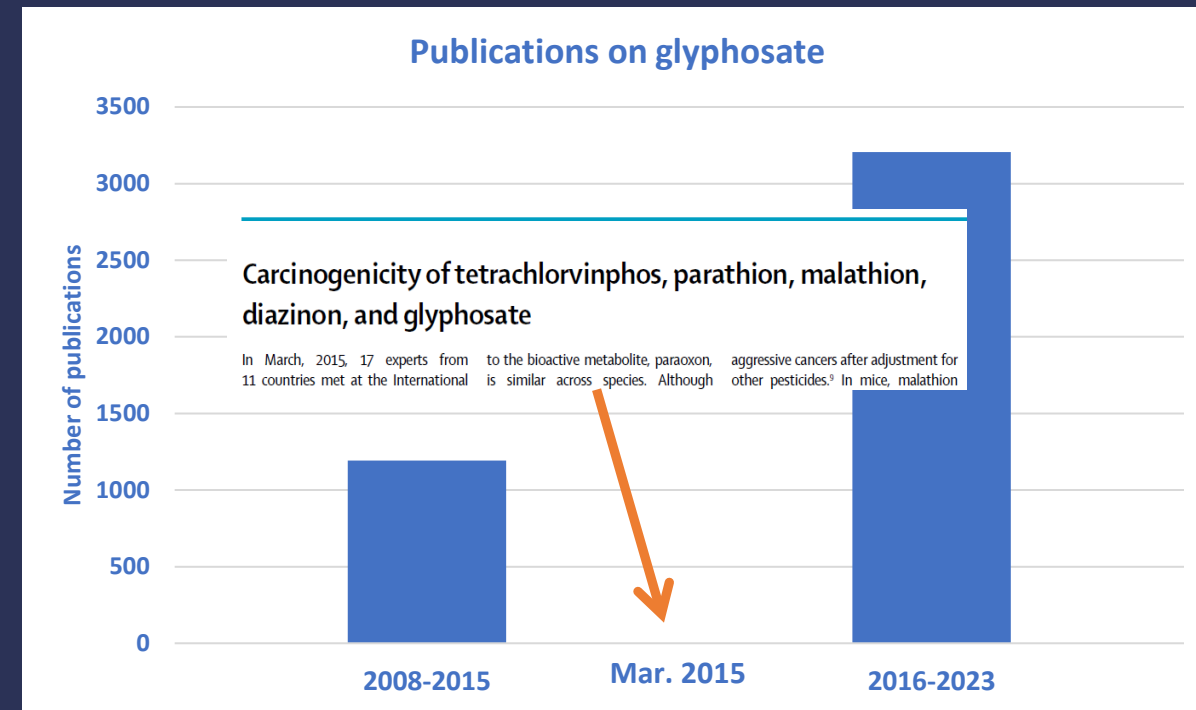


<https://handbooks.iarc.who.int>

Monographs : Dual role in global cancer research



- Evaluate results of latest scientific research in different disciplines
- Stimulate new research to fill identified gaps and to further develop findings



Presentation overview

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- Genomic epidemiology in the *IARC Monographs*



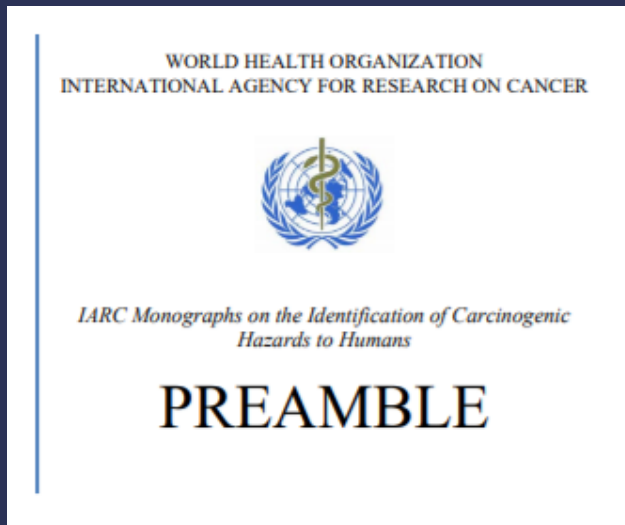
How are *IARC Monographs* evaluations conducted?



- Procedural guidelines for participant selection, conflict of interest, stakeholder involvement & meeting conduct
- Separate criteria for review of human, animal and mechanistic evidence
- Decision process for overall evaluations

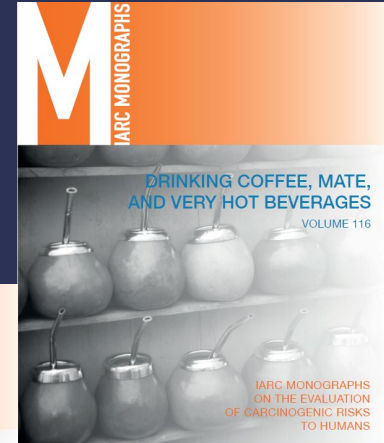
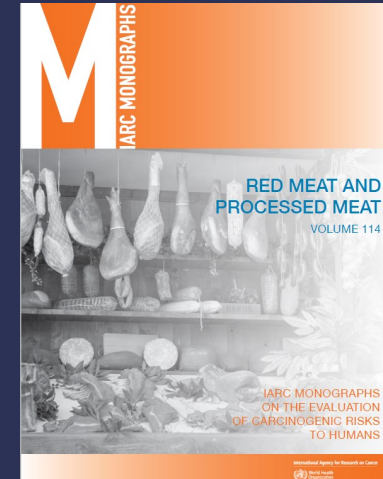
Highlights:

- Enhanced transparency
- Increased rigor of systematic review
- Modernized methods for literature searching and tagging

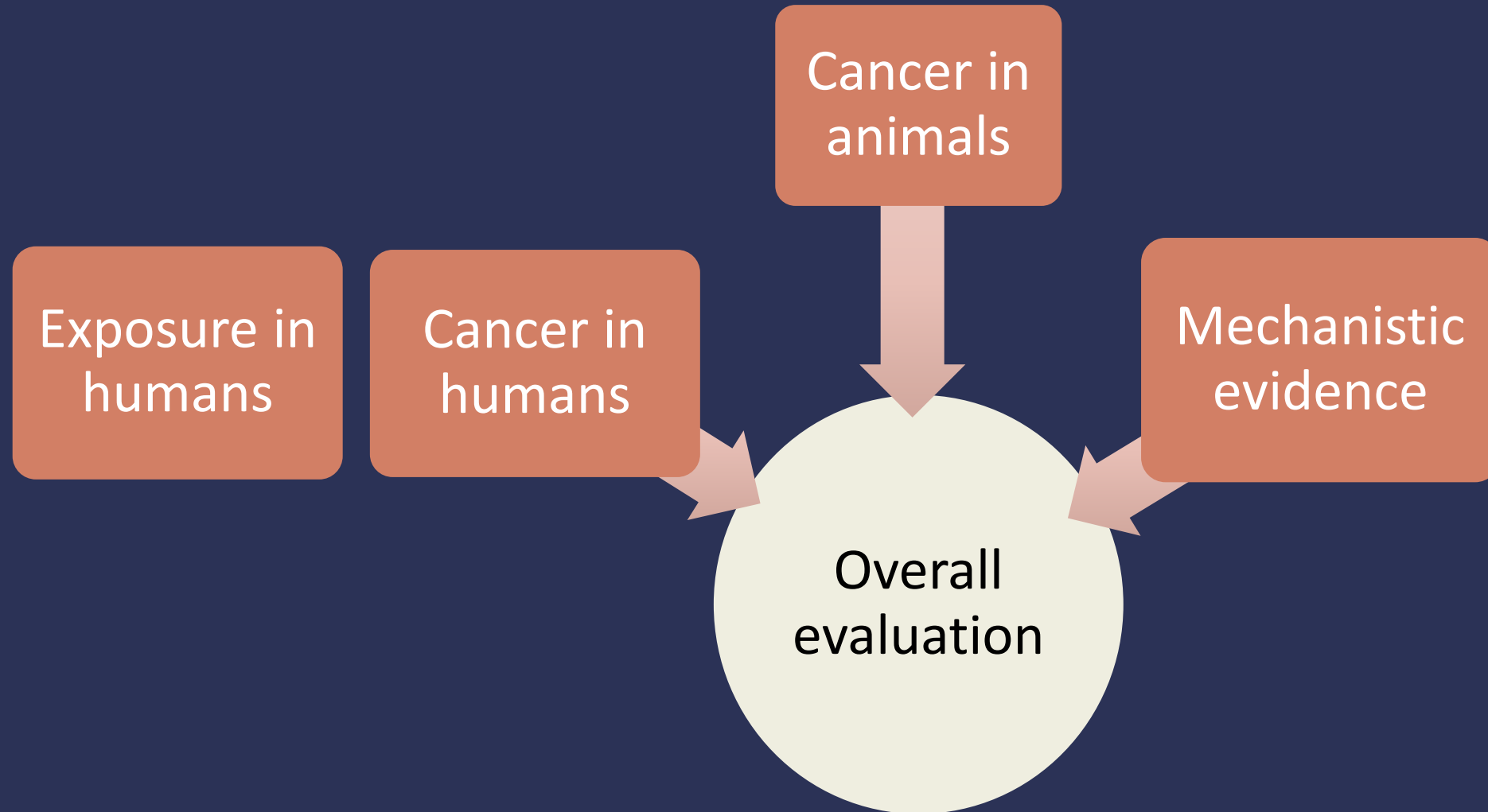


Preamble to the *IARC Monographs* (amended January 2019):

<https://monographs.iarc.who.int/wp-content/uploads/2019/07/Preamble-2019.pdf>



What evidence is considered?



Preamble to the *IARC Monographs* (amended January 2019): <https://monographs.iarc.who.int/wp-content/uploads/2019/07/Preamble-2019.pdf>

How is evidence evaluated?

Cancer in humans

— Preamble Part B, Section 6(a)

Cancer in experimental animals

Mechanistic evidence

Sufficient

- Causal relationship has been **established**
- Chance, bias, confounding could be **ruled out with reasonable confidence**

Limited

- Causal interpretation is **credible**
- Chance, bias, confounding **could not be ruled out with reasonable confidence**

Inadequate

- Studies permit **no conclusion** about a causal association, or
- **No data** were available

ESLC

- High-quality studies covering the full range of exposure are consistent in not showing a positive association at any level of exposure

How is evidence evaluated (*cont'd*)?

Cancer in humans	Cancer in Experimental Animals —Part B, Section 6(b)	Mechanistic evidence	
Sufficient	Limited	Inadequate	ESLC
• Causal relationship established: • <u>Multiple positive results</u> (↑Malignant tumours* in 2 species, OR studies, OR sexes of well-conducted [ideally GLP] study) • <u>Exceptionally, a single unusual result</u> (incidence, site/type, age, multi-site)	• Data suggest a carcinogenic effect but, for example: • Single study • ↑ Benign tumours only • ↑ Tumour multiplicity/ decr. latency only • Promoting activity only	• Studies permit no conclusion about the absence or presence of a carcinogenic effect, or • No data were available	• Well-conducted studies in at least two species show that the agent is not carcinogenic Conclusion is limited to the species, tumour sites, etc. (age of exposure, dose)
* or adequate combination of benign and malignant tumours			

Mechanistic data: *challenges*



IARC Monographs
Volume 100

- How to search systematically for relevant mechanisms?
- How to bring uniformity across assessments?
- How to analyze the voluminous mechanistic database efficiently?
- How to avoid bias towards favored mechanisms?

Ten key characteristics (KCs) of human carcinogens

Key characteristics:

1. Is electrophilic or can be metabolically activated
2. Is genotoxic
3. Alters DNA repair or causes genomic instability
4. Induces epigenetic alterations
5. Induces oxidative stress
6. Induces chronic inflammation
7. Is immunosuppressive
8. Modulates receptor-mediated effects
9. Causes immortalization
10. Alters cell proliferation, cell death, or nutrient supply

- Chemical and biological properties of **established** human carcinogens
- Data on key characteristics can provide evidence of carcinogenicity
- Used to assemble data relevant to mechanisms of carcinogens— without needing an *a priori* hypothesis of the mechanism

- Smith MT, Guyton KZ, Gibbons CF, Fritz JM, Portier CJ, Rusyn I, DeMarini DM, Caldwell JC, Kavlock RJ, Lambert PF, Hecht SS, Bucher JR, Stewart BW, Baan RA, Coglian VJ, Straif K (2016); Key characteristics of carcinogens as a basis for organizing data on mechanisms of carcinogenesis. *Env Health Persp.*, 124(6):713-21
- Guyton KZ, Rusyn I, Chiu WA, Corpet DE, van den Berg, M, Ross, M, Christiani DC, Beland FA, Smith MT (2018); Application of the key characteristics of carcinogens in cancer hazard identification. *Carcinogenesis*, 39(4):614
- IARC Scientific Publication No. 165: Tumour Site Concordance and Mechanisms of Carcinogenesis (2019). <https://publications.iarc.fr/578>
- Smith MT, Guyton KZ (2020). Identifying carcinogens from 10 key characteristics: a new approach based on mechanisms. In: Wild CP, Weiderpass E, Stewart BW, editors. World Cancer Report: Cancer Research for Cancer Prevention. <https://publications.iarc.fr/586>

How is evidence evaluated (*cont'd*)?

Cancer in humans	Cancer in experimental animals	Mechanistic evidence —Part B, Section 6(c)
<i>Strong</i>	<i>Limited</i>	<i>Inadequate</i>
• Results in different systems are consistent • Overall database is coherent • Studies cover a range of relevant end-points in mammalian species	• Data are suggestive but, for example: • Studies cover a narrow range of experiments, end-points or species • Unexplained incoherence across studies of similar design, or different end-points and test systems	• Few or no data were available • Questions remain about the adequacy of the design, conduct, interpretation of the studies • Available results are negative

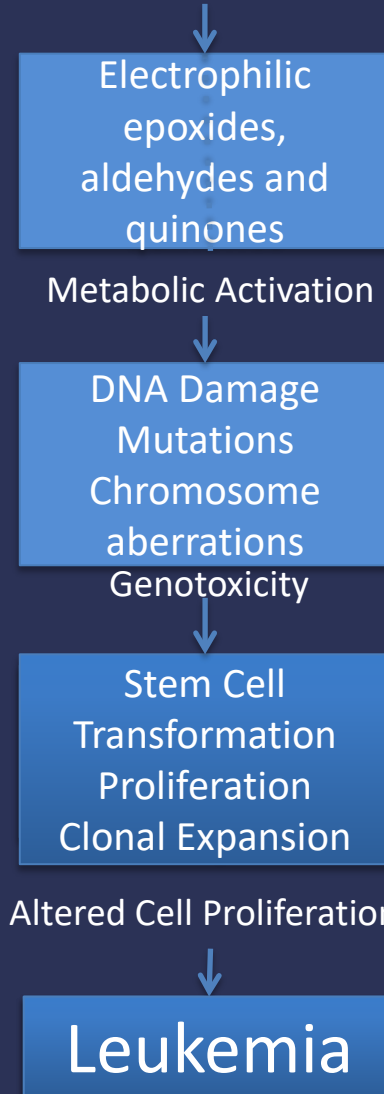
Three distinct mechanistic topics can influence overall evaluations

Cancer in humans	Cancer in experimental animals	Mechanistic evidence —Part B, Section 6(c)
<i>Key Characteristics of Carcinogens</i>	<i>Mechanistic Class</i>	<i>Mechanism not relevant</i>
• In exposed humans • In human primary cells or tissues • In experimental systems	• Belongs, based on mechanistic considerations, to a class of agents— one or more of which is probably carcinogenic or carcinogenic to humans	• Meets criteria for concluding that mechanism of carcinogenicity in experimental animals is not relevant to humans

Key characteristics of benzene

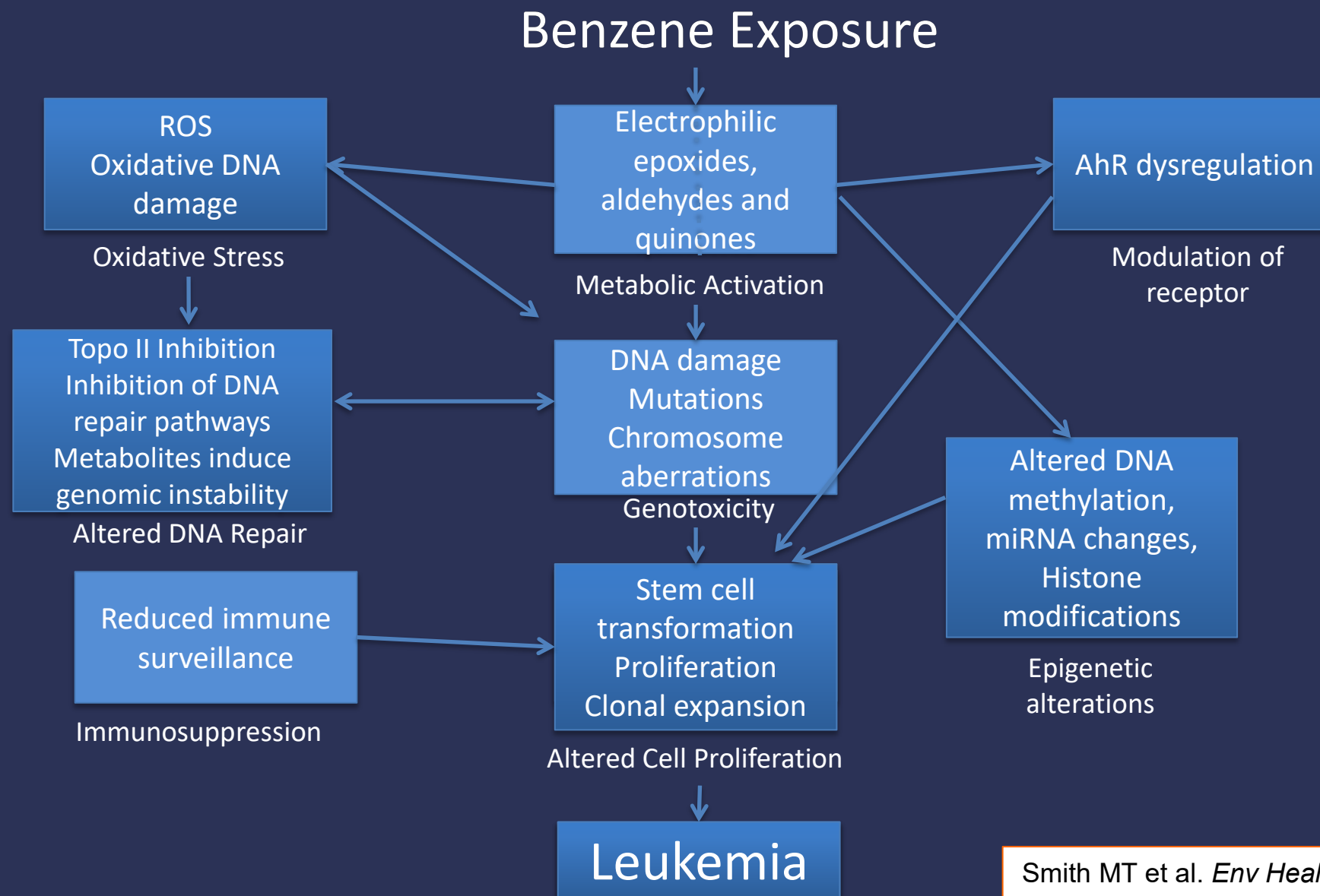
Benzene Exposure

An Adverse Outcome
Pathway?



An Adverse Outcome Network

Involving 8 key characteristics of human carcinogens



End-point measurement and integration of the KCs for cancer hazard identification: IARC Workshop report

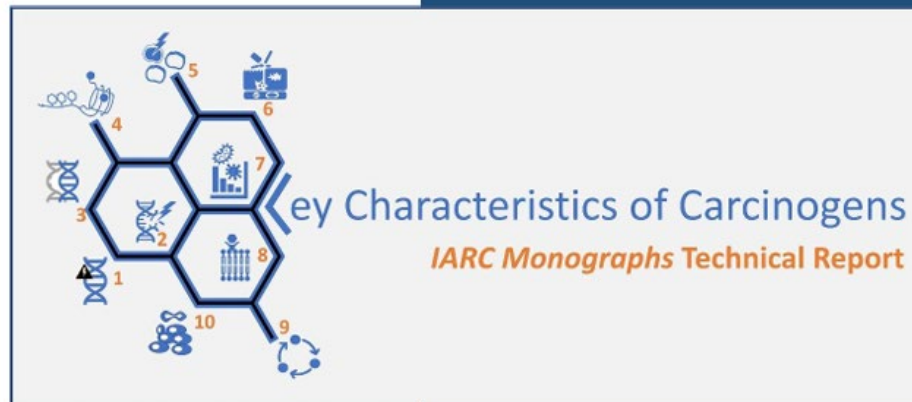
International Agency for Research on Cancer



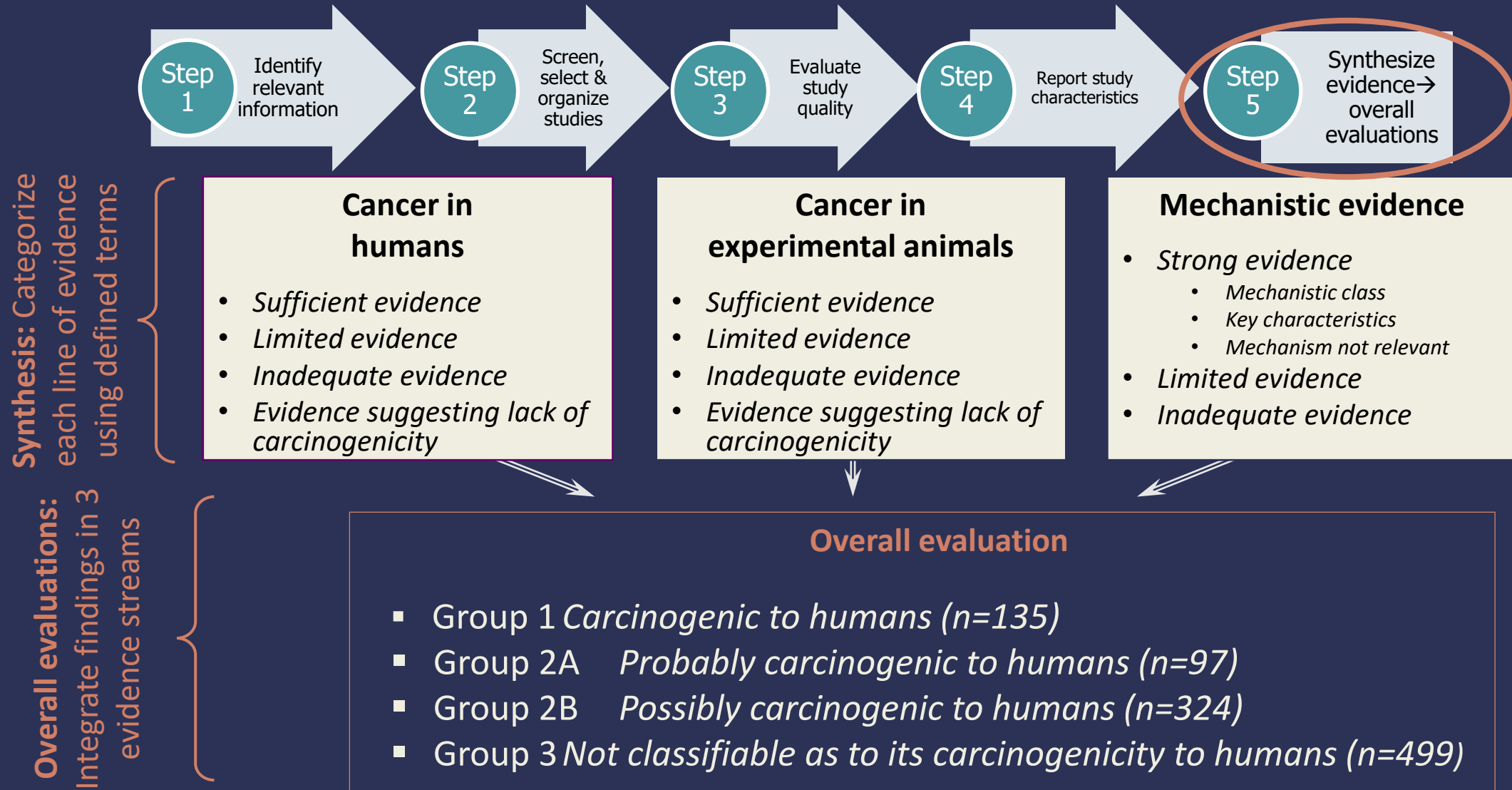
IARC Monographs Programme

*Key Characteristics-associated End-points for Evaluating
Mechanistic Evidence of Carcinogenic Hazards*

<https://monographs.iarc.who.int/wp-content/uploads/2025/06/KCW-FINAL.pdf>



Evidence synthesis and integration



Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited		Strong	
		Strong (human cells or tissues)	
		Strong (mechanistic class)	Possibly carcinogenic (Group 2B)
Limited			
	Sufficient		
		Strong	Not classifiable (Group 3)
	Sufficient	Strong (does not operate in humans)	
All other situations not listed above			

Sufficient for cancer in humans

Recent classifications: 2019 Preamble

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		
Limited			Probably carcinogenic (Group 2A)
Limited			Possibly carcinogenic (Group 2B)
	Sufficient	Strong (does not operate in humans)	Not classifiable (Group 3)
All other situations not listed above			

Opium consumption: *Sufficient evidence* for cancers of larynx, lung & bladder. *Limited evidence* for cancers of pharynx, oesophagus, stomach, pancreas.

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited		Strong	
	Sufficient	Strong (human cells or	
			Carcinogenic (Group 2B)
Limited			
	Sufficient		
		Strong	Not classifiable (Group 3)
	Sufficient	Strong (does not operate in humans)	
All other situations not listed above			

Sufficient for cancer in animals + strong KCs in exposed humans

Recent classifications: 2019 Preamble

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient	Sufficient	Strong (exposed humans)	Carcinogenic (Group 1)
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited			Possibly carcinogenic (Group 2B)
Limited			Not classifiable (Group 3)
	Sufficient	Strong (does not operate in humans)	
All other situations not listed above			

PFOA: *Sufficient evidence* for cancer in experimental animals. *Strong evidence* for KCs 4 & 7 in exposed humans. (Also, *limited evidence* for cancers of kidney & testes in humans, *strong evidence* for KCs in experimental systems)

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited		Strong	
	Sufficient	Strong (human cells or tissues)	
		Strong (mechanistic class)	
Limited			
	Sufficient		
		Strong	
	Sufficient	Strong (does in hum	
All other situations not listed above			

Limited cancer in humans + either (or both) *Sufficient* cancer in animals or *Strong* mechanistic evidence

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited		Strong	
	Sufficient	Strong (human cells or tissues)	
		Strong (mechanistic class)	
Limited			
	Sufficient		
		Strong	
	Sufficient	Strong (does in hum	
All other situations not listed above			

Sufficient cancer in animals and Strong mechanistic evidence in primary human cells or tissues

Sufficient cancer in animals and *Strong* mechanistic evidence in primary human cells or tissues

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited		Strong	
	Sufficient	Strong (human cells or tissues)	
		Strong (mechanistic class)	
Limited			Possibly carcinogenic (Group 2B)
	Sufficient		
		Strong	Not classifiable (Group 3)
	Sufficient	Strong (does not operate in humans)	
All other situations not listed above			

Strong mechanistic class

*Strong
mechanistic class*

Recent classifications: 2019 Preamble

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited		Strong	
	Sufficient	Strong (human cells or tissues)	
		Strong (mechanistic class)	
Limited			Possibly carcinogenic (Group 2B)
	Sufficient		
			Not classifiable (Group 3)

- **Aromatic amines** (aniline, *ortho*-anisidine)
- **Epoxide** (Glycidyl methacrylate)

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient	Sufficient	Sufficient (in humans)	Carcinogenic (Group 1)
Limited	Sufficient	Sufficient (in humans)	Probably carcinogenic (Group 2A)
Limited	Sufficient	Sufficient (in humans)	
	Sufficient	Sufficient (in humans)	
	Sufficient	Sufficient (in humans)	Probably carcinogenic (Group 2A)
	Sufficient	Sufficient (in humans)	
	Sufficient	Sufficient (in humans)	
Limited	Sufficient	Strong (mechanistic class)	Possibly carcinogenic (Group 2B)
	Sufficient	Strong	Possibly carcinogenic (Group 2B)
	Sufficient	Strong (does not operate in humans)	Not classifiable (Group 3)
All other situations not listed above			Not classifiable (Group 3)

Limited for human cancer alone

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited			
	Sufficient	or	
		Strong (mechanistic class)	
Limited			Possibly carcinogenic (Group 2B)
	Sufficient		
		Strong	
	Sufficient	Strong (does not operate in humans)	Not classifiable (Group 3)
All other situations not listed above			

Sufficient for cancer in animals alone

Recent classifications: 2019 Preamble

There are more than 250 agents classified with *sufficient* evidence for cancer in experimental animals and *inadequate* evidence regarding cancer in humans. These could benefit most from studies of mechanistic evidence in exposed humans, human (or humanized) primary cells/tissues. Examples:

- **Acrylamide**
- Multiwalled carbon nanotubes
- Carbon black & titanium dioxide
- 1-bromopropane
- Individual dyes & PAHs
 - Aniline
 - Dyes metabolized to benzidine

Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient	Strong (exposed humans)	Carcinogenic (Group 1)
Sufficient	Strong	Probably carcinogenic (Group 2A)
Sufficient	Strong (human cells or tissues)	
	Strong (mechanistic class)	
Sufficient		Possibly carcinogenic (Group 2B)
	Strong	
Sufficient	Strong (does not operate in humans)	Not classifiable (Group 3)
All other situations not listed above		

Overall evidence classification

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
Limited			
	Sufficient		
			Possibly carcinogenic (Group 2B)
Limited			
	Sufficient		
		Strong	
	Sufficient	Strong (does not operate in humans)	Not classifiable (Group 3)
All other situations not listed above			

Strong for
mechanistic
evidence alone

Recent classifications: 2019 Preamble

Evidence of Cancer in Humans	Evidence of Cancer in Experimental Animals	Mechanistic Evidence	Evaluation
Sufficient			Carcinogenic (Group 1)
	Sufficient	Strong (exposed humans)	
Limited	Sufficient		Probably carcinogenic (Group 2A)
		Strong	
	Sufficient	Strong (human cells or tissues)	
		Strong (mechanistic class)	Possibly carcinogenic (Group 2B)
		Strong	
	Sufficient	Strong (does not operate in humans)	Not classifiable (Group 3)
All other situations not listed above			

Mechanistic alone: aldehyde line

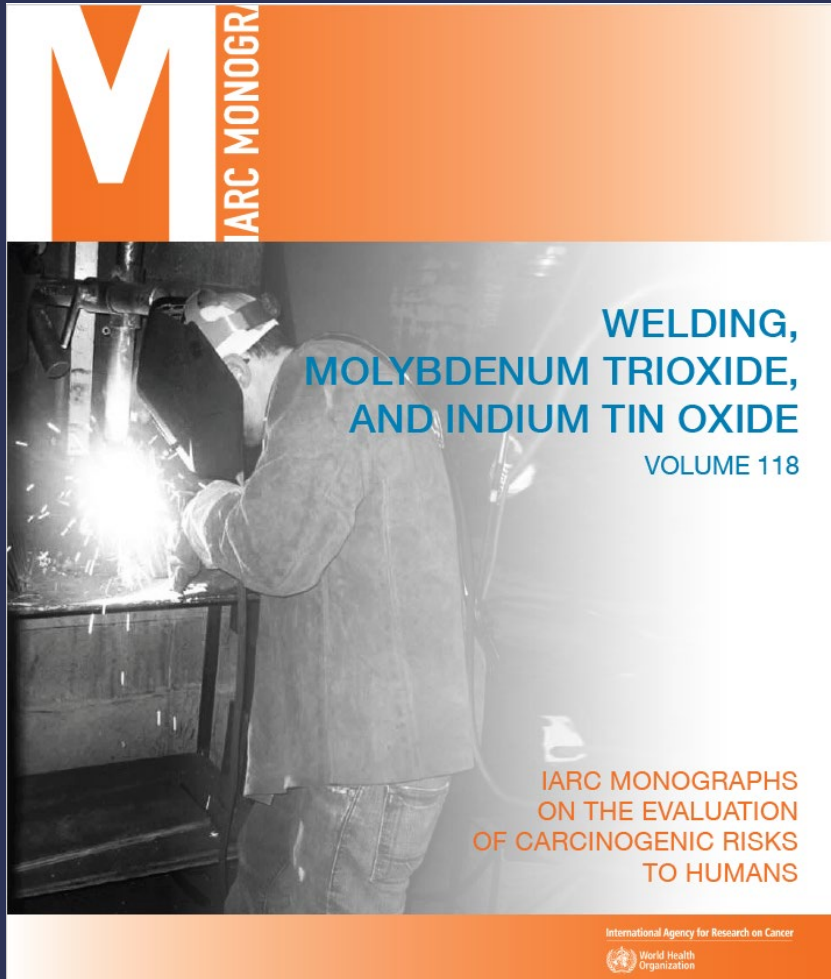
Strong mechanistic evidence alone:

- Crotonaldehyde
- Arecoline
- PFOS

Recent mechanistic evaluations in humans

Volume (year)	Agent	Evaluation of human evidence	Key characteristics observed in humans
132 (2022)	Firefighting exposure	Strong	Genotoxicity, epigenetic effects, oxidative stress, chronic inflammation, receptor-mediated effects
133 (2023)	2-bromopropane	Suggestive	Immunosuppression, receptor-mediated effects
135 (2023)	PFOA, PFOS	Strong	Epigenetic effects, immunosuppression
136 (2024)	Talc	Suggestive	Chronic inflammation, alters cell proliferation
137 (2024)	Tacrolimus	Strong	Immunosuppression
137 (2024)	Voriconazole + UVR	Strong	Alters cell proliferation
138 (2025)	Automotive gasoline	Strong	Genotoxicity, oxidative stress, chronic inflammation
139 (2025)	Merkel cell polyomavirus	Strong	Genotoxicity
139 (2025)	Hepatitis D virus	Strong	Chronic inflammation

How can mechanistic studies in humans be more informative?



- Features of study population?
- Attributes of study design?
 - Adequate power, minimal selection bias and residual confounding
- Attributes of endpoint selection and measurement?
 - How to measure complex endpoints? (immunosuppression, chronic inflammation)
 - Persistence & specificity of KC endpoint (e.g., electrophilicity & genotoxicity vs. epigenetic effects & oxidative stress)
- Attributes & quality of exposure assessment?
- Quantity of studies?

Presentation overview

- IARC and the *Monographs*- background
- The *Monographs* Preamble – modernized evidence synthesis and integration across evidence streams
- Priorities for *IARC Monographs* evaluation
- How has genomic epidemiology been used in the *IARC Monographs*?



Priorities recommended for evaluation by the IARC Monographs during 2025-2029



International Agency for Research on Cancer



monographs.
iarc.who.int

ACKNOWLEDGEMENTS

Financial support for the Monographs was received from:

- National Cancer Institute, USA (Cooperative Agreement U01 CA33193)
- US NIEHS/National Toxicology Program
- European Commission (DG for Employment, Social Affairs, and Inclusion; and EaSI (<http://ec.europa.eu/social/easi>))

Staff of the IARC Monographs on the Identification of Carcinogenic Hazards to Humans



IARC (2024). Report of the Advisory Group to Recommend Priorities for the IARC Monographs during 2025–2029. IARC Monographs on the Identification of Carcinogenic Hazards to Humans. [in preparation]

Advisory Group on Priorities March 2024

Streams of evidence

Red Mechanistic evidence

Blue Human cancer

Yellow Cancer bioassay

Green Bioassay + human cancer

Orange Mechanistic + bioassay

Purple Mechanistic + human cancer

Black Mechanistic + human + bioassay

Agents not previously evaluated by IARC
are in **bold type**

High priority agents (< 2.5 or < 5 years) are
placed within the inner yellow ring;
Medium priority agents are placed outside
the yellow ring

[Agent]*: High priority (< 5 years)

Berrington de González et al. (2024). *Lancet Oncol.*
[Published online 12 April 2024](#)



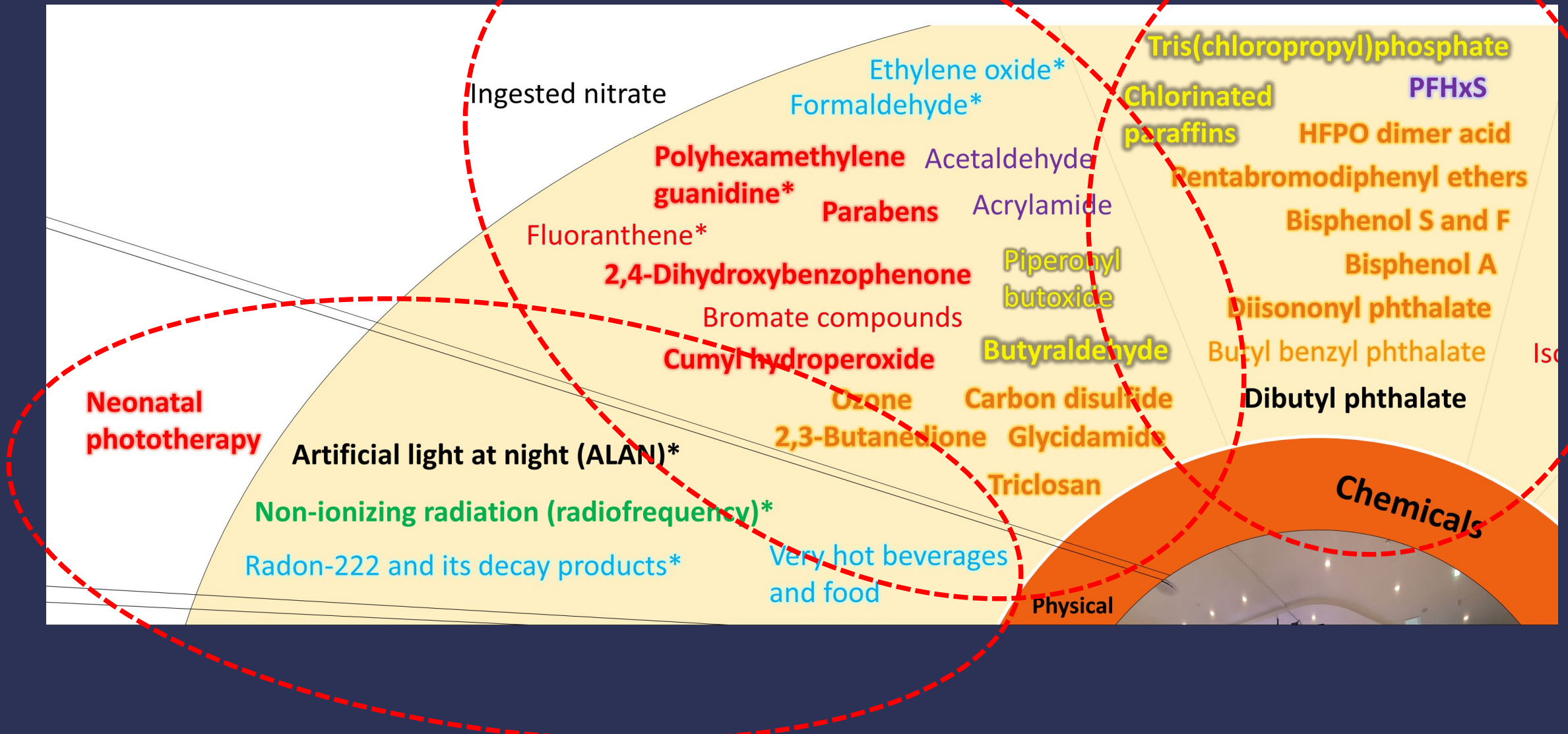
Advisory Group to Recommend Priorities for the
IARC Monographs during 2025–2029
IARC, France, 19–22 March 2024

International Agency
for Research on Cancer

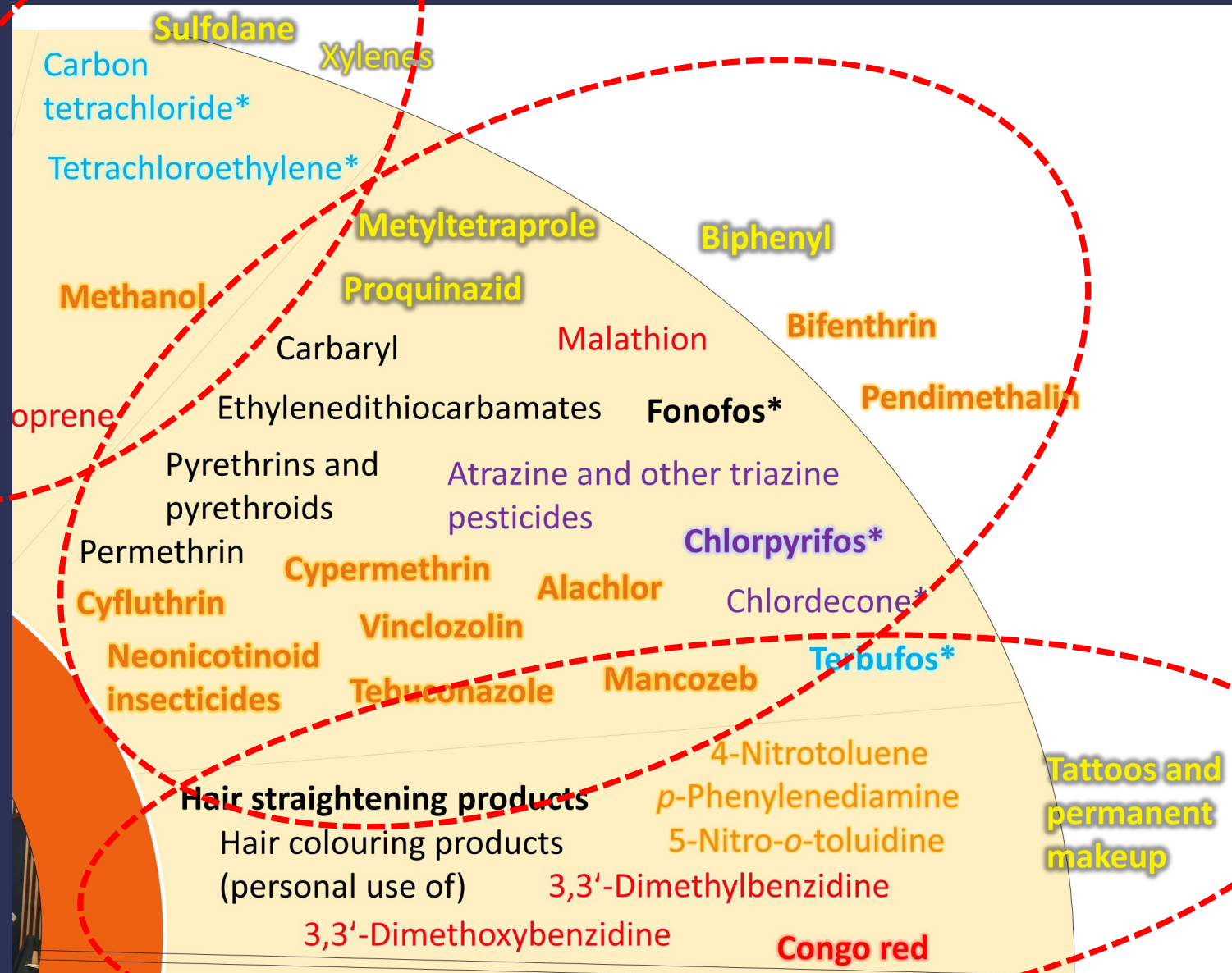


28 scientists from 22 countries,
free from conflicts of interest

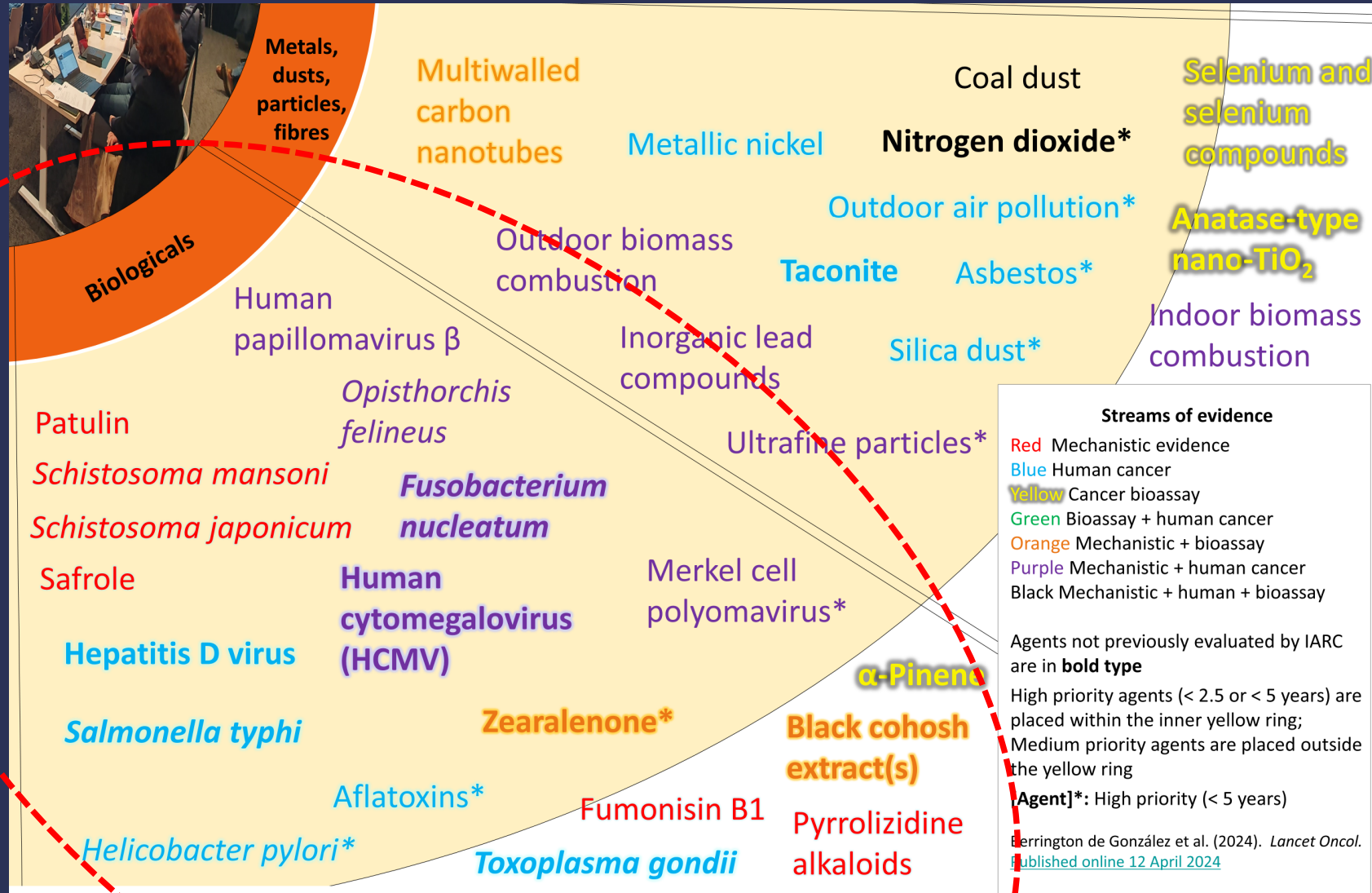
POPs, miscellaneous chemicals, physical agents



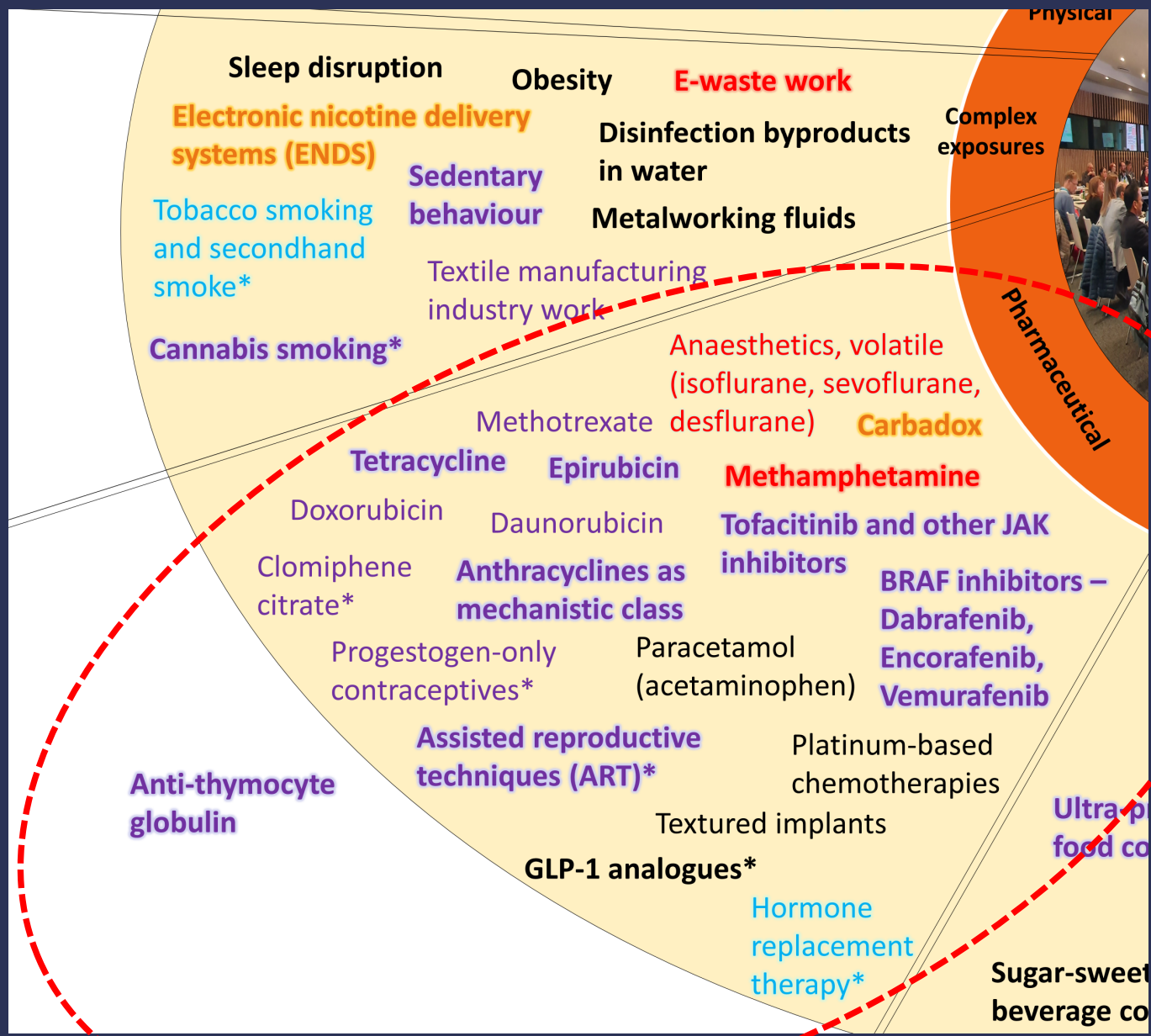
Solvents, pesticides, dye-related



Biological agents



Pharmaceuticals



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- Genomic epidemiology in the *IARC Monographs*



Genomic studies have been influential

- Since Volume 100 (2009), genomic considerations have been important for evaluating at least 35 agents
- Many of these studies involve p53 mutations, mutational signatures, and/or polymorphisms in metabolizing enzymes

Genomic studies have been influential

Volume	Agent/s	Mention of specific polymorphisms, alleles, or mutational signatures
100A	aristolochic acid	molecular signatures in the DNA (e.g. the TP53 mutation signature)
100A	tamoxifen	G→T transversions.
100A	etoposide	high frequency of 11q23 translocations in the leukaemias
100B	O. viverrini-endemic areas	Differences in Ki-RAS mutational status have been described when comparing cholangiocarcinoma from Japanese patients.
100C	Silica dust, crystalline (quartz or cristobalite)	Specific mutations in KRAS and p53, specific to various codons and linked to silicosis
100D	solar and UV radiation	Specific polymorphisms were reported for TNF-alfa and IL-1RA these are people at high risk of silicosis
100D	X- and γ-radiation	Specific mutations in p53 and H-Ras genes in rodents, similar mutations observed in patients with leukaemia.
100E	Alcohol cons. (EtOH, Acetaldehyde)	Polymorphisms in ADH and ALDH metabolizing enzymes, differences in head and neck cancers
100E	nitroso compounds class	specific mutational signature
100E	tobacco smoking	specific mutational signature in p53 and KRAS in various tissues
100E	NNN-NNK	Human cancer genomes harbour the mutational signature of tobacco-specific nitrosamines NNN and NNK
100F	ethylene oxide	specific mutations in p53 and H-Ras genes in rodents, similar mutations observed in patients with leukaemia.
100F	1,3-butadiene metabolites: 1,2-epoxybutene (EB) and 1,2,3,4-diepoxybutane (DEB)	specific mutational spectra in all the systems
100F	Benzo[a]pyrene	Benzo[a]pyrene-induced skin tumours harboured G→T transversion mutations in the TP53 tumour-suppressor gene. The specific DNA adducts occurred at guanine positions in codons 157, 248, and 273 of the TP53 gene in anti-benzo[a]pyrene-7,8-diol-9,10-epoxide-treated human HeLa cells. The same positions are the major mutational hotspots found in human lung cancers
100F	4-Aminobiphenyl	Specific mutations at HPRT locus. In human bladder cells treated with N-hydroxy-4-ABP, preferential sites of adduct formation in TP53 were at codons 175, 248, 280, and 285, which are mutational hotspots for cancer of the urinary bladder
101	Di(2-ethylhexyl) phthalate	Variants of PPAR alpha in different populations.
103	some N- and S-heterocyclic PAHs, dibenz(c,h) acridine	A to T and G to T transversion mutations in H-Ras gene consistent with DNA binding to dG or dA.
104	malaria virus	various HLA polymorphisms, cytokines polymorphisms
105	Diesel and gasoline engine exha.	Signature metabolites and signature DNA adducts arising from the three pathways of benzo[a]pyrene activation.
106	dichloroacetic acid	GSTZ1 polymorphic alleles
106	trichloroethylene	GSTT1a allele; CYP2E1 variant alleles
107	Polychlorinated biphenyls	various polymorphisms in several hormone-metabolizing genes and susceptibility to cancer of the testis
108	outdoor air pollution	Polymorphisms in metabolizing enzymes CYPs, GST
112	diazinon	It was hypothesized that the PON1 Q192R polymorphism can influence susceptibility to organophosphates
114	Red meat and processed meat	In a nested case-control study in the same cohort, a statistically significant increase in colorectal cancer risk per red meat intake was observed among carriers of the homozygous variant XPC Lys939Gln

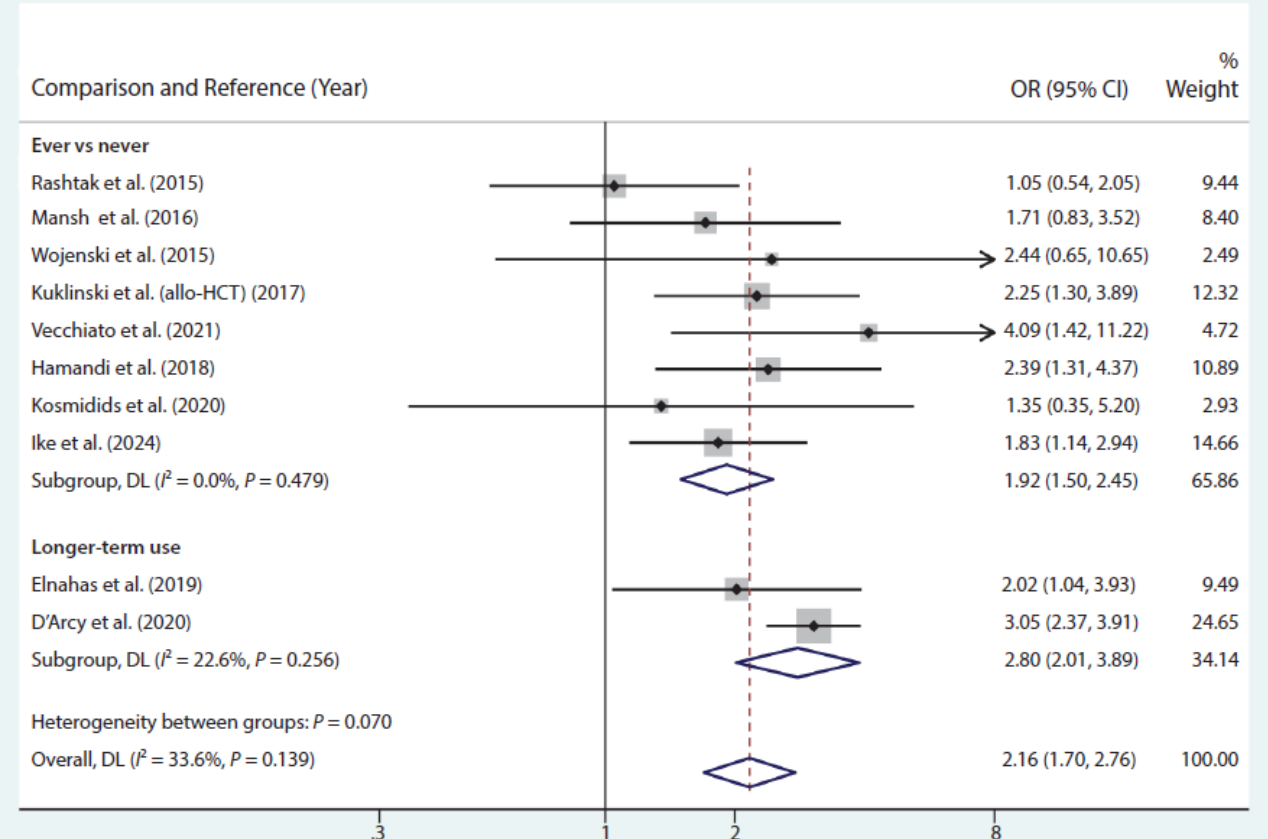
Genomic studies have been influential

Volume	Agent/s	Mention of specific polymorphisms, alleles, or mutational signatures
115	hydrazine	Hydrazine was metabolized to acetylhydrazine (monoacetylhydrazine) in exposed workers. In male Japanese workers, the rate of acetylation was dependent on polymorphisms in N-acetyltransferase (NAT2), with half-lives of about 2 and 4 hours in rapid and slow acetylators, respectively
115	N,N-dimethylformamide	CYP and GST alleles and cancer susceptibility (vol 115, section 4.4)
116	coffee	Coffee and/or caffeine preference is a highly heritable trait. Several large-scale genome-wide association studies (GWAS) and meta-analyses point to a small number of alleles, most notably polymorphisms in AHR and CYP1A genes, that are very strongly and consistently associated with the patterns of coffee consumption.
117	aldrin	Men carrying two AA alleles at rs7679673 in TET2 were at increased risk of prostate cancer when they had high aldrin use.
120	benzene	Individuals carrying a homozygous null polymorphism for the quinone-metabolizing enzyme NQO1, and therefore lacking the ability to detoxify electrophilic quinones, were more susceptible to benzene toxicity. Also, other enzymes
121	styrene	Urinary concentrations of corresponding metabolites have been associated with these genetic polymorphisms. Decreased excretion levels of urinary MA and PGA, as well as decreased expression of CYP2E1 mRNA levels, were noted in 49 subjects exposed to styrene carrying the heterozygous allele of CYP2E1*5B (c1/c2) when compared with subjects carrying the homozygote (c1/c1).
124	night shift work	Seven polymorphisms in clock genes, as well as genes involved in melatonin metabolism, and night shift work. The study found some suggestive evidence for interactions primarily for a CLOCK single-nucleotide polymorphism (SNP) (rs10462028), with an increased risk of cancer of the breast among shift workers for AA versus GG.
131	cobalt metal	Mutational spectra of lung tumours induced by cobalt metal in the NTP (2014) studies in mice and rats showed mutations (especially G→T transversions) in K-Ras, Tp53, and Egfr genes compatible with oxidative DNA damage.
132	occupational exposure as firefighter	Most common COSMIC mutation signatures observed in the WTC-exposed firefighters were also for (1) ageing; (2) DNA mismatch repair; (3) smoking; and (4) alkylating agents
137	voriconazole	CYP19A1 alleles- fast and slow metabolizers
137	tacrolimus	CYP3A5*6 and *7 alleles - fast and slow metabolizers and lower immunosuppressive function
139	Merkell cell virus	specific signature in sT and LT sequences

Recent example from v.137: Voriconazole

- Most were traditional cohort or nested case-control studies of skin SCC from use of antifungal voriconazole in transplant patients
- Ever-use: metaOR=1.9 (95% CI: 1.5-2.4)
- Long-term use: metaOR=2.8 (95% CI: 2.0-3.9)

Fig. 2.3 Forest plot for random-effects meta-analysis of the risk of squamous cell carcinoma of the skin among users of voriconazole, stratified by measure of use analysed

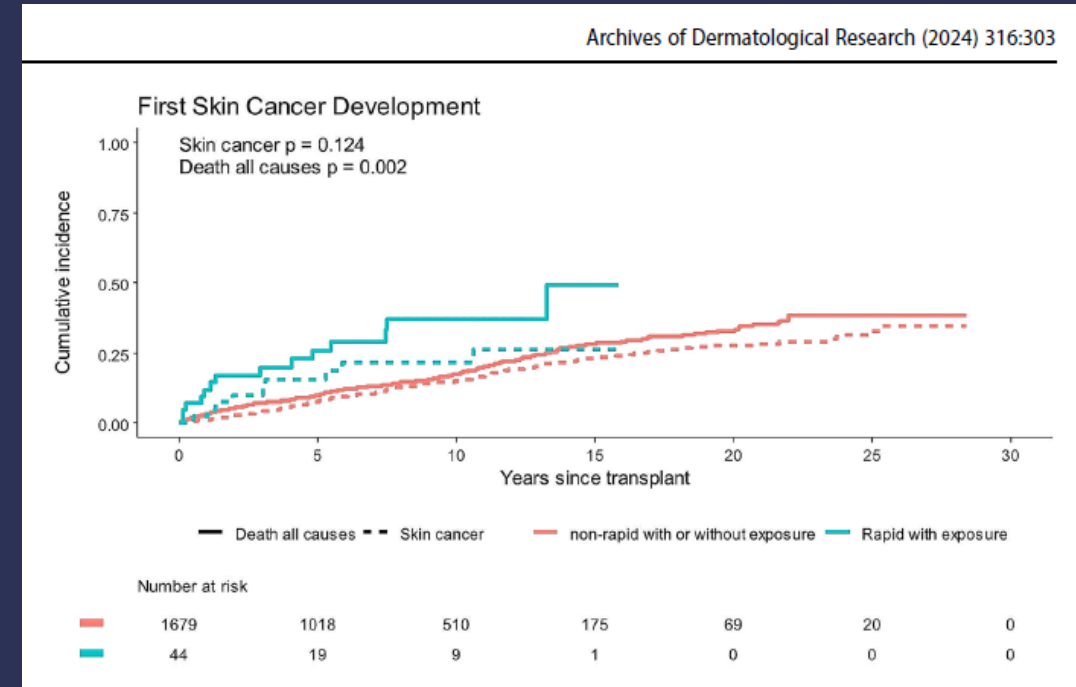


NOTE: Weights and between-subgroup heterogeneity test are from random-effects model

<https://publications.iarc.who.int/659>

Recent example from v.137: Voriconazole

- Two studies on the ultrarapid metabolizing CYP2C19*17 allele and risk of SCC in transplant patients
- Shorter time-to-first-tumour (and higher total number) in ultra-rapid metabolizers
- Risk associated with voriconazole seemed explained by metabolizer status (Williams and Arron 2016).
- The real actor appears to be the *N*-oxide-voriconazole



Ike et al. 2024

Studies had limitations (e.g., not restricted to only voriconazole users)

Recent example from v.137: Voriconazole

- Case reports of early-onset SCC of skin among children who were transplant recipients
- Despite concerns about potential biases away from the null in the cohort studies, the evidence overall supported a causal interpretation for voriconazole and SCC
- Strong mechanistic evidence (cell proliferation—actinic keratosis)

Published in final edited form as:

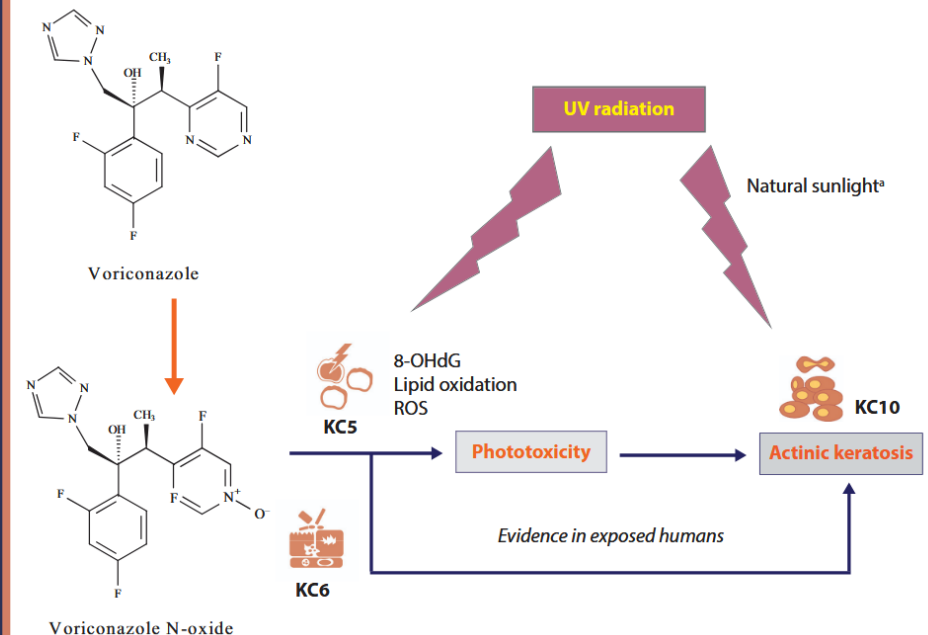
J Am Acad Dermatol. 2010 January ; 62(1): 31. doi:10.1016/j.jaad.2009.09.033.

Chronic Phototoxicity and Aggressive Squamous Cell Carcinoma of the Skin in Children and Adults During Treatment with Voriconazole

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Fig. 4.2 Induction of actinic keratosis in exposed humans



Synthesizing body of human cancer evidence

An adaptation of Bradford Hill viewpoints

- Strength of association (but weak associations may be causal)
- Consistency of association, accounting for study quality and informativeness
- Presence of exposure-response
- Temporality of association
- Biological plausibility

Evidence triangulation across studies with different types of biases



Approaches to assess impact of biases

- Explicit exposure quality evaluations
- Triangulation of evidence
- Use of causal diagrams to better elucidate confounders
- Recognizing trade-offs of different study designs for control of bias
- Negative control outcomes
- Evaluation of indirect adjustments for confounders
- Theoretical adjustment based on worst-case assumptions about confounder-exposure distributions

Different study designs have different biases

Anticipated impact on measures of association

No need to adjust for non-confounders

e.g., Mendelian randomization
(may be useful in some settings)

Outcomes plausibly related to confounders but not agent

Useful when information on confounder available for no or only some subjects

Conclusions

- The *IARC Monographs* programme has a long history of evidence synthesis and integration (cancer epidemiology, toxicology, mechanisms), which continues to evolve with scientific advances in epidemiology and toxicology
- Genotoxicity is an important key characteristic of carcinogens, and information on specific mutations, signatures or gene polymorphisms can powerfully affect carcinogenicity classification if found in exposed humans
- Genomic epidemiology often supports causal conclusions through triangulation with other types of evidence

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